Xinyu Shi

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RESEARCH INTERESTS

Human-Computer Interaction · Human-Centered AI

EDUCATION

2021 – Present **Ph.D. in Computer Science**
University of Waterloo, Canada
Advisor: Prof. Jian Zhao

2016 – 2020 **B.Eng. in Software Engineering**
Xiamen University, China
Advisor: Prof. Shihui Guo

EMPLOYMENT

Summer 2025 **Adobe Research** — Creative Technologies Lab
Research Intern
Hosts: Rubaiat Habib Kazi, Li-Yi Wei, Cherry Zhao
Research on redefining how creators interact with and control VLMs in animation creation [P10].

Summer 2024 **Adobe Research** — AI Experience Lab
Research Intern
Hosts: Shunan Guo, Jane Hoffswell, Gromit Chan, Eunyee Koh
Research on computational representations of abstract intent for infographic creation [A1].

Summer 2023 **Microsoft Research** — Data, Knowledge & Intelligence Group
Research Intern
Host: Yun Wang
Research on repurposing visualizations as instruments for video color authoring [P6].

AWARDS

2026 **Best Paper Honorable Mention Award** at *ACM CHI* as first author
2025 **Spotlight** at *NeurIPS MTI-LLM Workshop* as co-author
2024 **Best Paper Award** at *ACM CHI* as first author
2022 **Oral Recognition** at *ACM MM* as co-first author
2026 **Jane Street Fellowship, Honorable Mention**
2025 – 2027 **David R. Cheriton Graduate Scholarship**, University of Waterloo
2021 – 2026 **International Doctoral Student Award**, University of Waterloo
2021 **Vector Scholarship in AI**, Vector Institute
2016 – 2020 **Award for Excellence in Academic Performance**, Xiamen University

PEER-REVIEWED PUBLICATIONS (FULL PAPERS)

Google Scholar: [link]; *Semantic Scholar:* [link]

Note about venues: *CHI* is considered as the top-tier, most prestigious venues in the field of Human-Computer Interaction (HCI), with an annual acceptance rate of 20 - 25%.

∗ denotes Equal Contribution.

[P10] CHI 2026 **Xinyu Shi**, Li-Yi Wei, Nanxuan Zhao, Jian Zhao, Rubaiat Habib Kazi. Notational Animating: An Interactive Approach to Creating and Editing Animation Keyframes. *In Proceedings of the ACM Human Factors in Computing Systems (CHI). 2026.* DOI
Best Paper Honorable Mention.

[P9] ACL 2026 Zijian Chen*, Xueguang Ma*, Shengyao Zhuang*, Ping Nie, Kai Zou, Andrew Liu, Joshua Green, Kshama Patel, Ruoxi Meng, Mingyi Su, Sahel Sharifymoghaddam, Yanxi Li, Haoran Hong, **Xinyu Shi**, Xuye Liu, Nandan Thakur, Crystina Zhang, Luyu Gao, Wenhu Chen, Jimmy Lin. BrowseComp-Plus: A Fair and Disentangled Evaluation Benchmark for Deep Search Agents. *In Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL). 2026.* DOI
Spotlight at NeurIPS 2025 *MTI-LLM* Workshop.
Adopted by *Anthropic (Claude Opus 4.5 launch), Databricks, Microsoft, Google, etc.*

[P8] SIGIR-AP 2026 Crystina Zhang, Zigeng Xu, **Xinyu Shi**, Raphael Tang, Nandan Thakur, Xueguang Ma, Yao Lu, Jimmy Lin. Hawkeye: Inspecting Agentic Search Trajectories. *In Proceedings of the ACM SIGIR Conference on Information Retrieval in the Asia Pacific. 2026. (in press)*

[P7] CHI 2025 **Xinyu Shi**, Yinghou Wang, Ryan Rossi, Jian Zhao. Brickify: Enabling Expressive Design Intent Specification through Direct Manipulation on Design Tokens. *In Proceedings of the ACM Human Factors in Computing Systems (CHI). 2025.* DOI

[P6] CHI 2024 **Xinyu Shi**, Yinghou Wang, Yun Wang, Jian Zhao. Piet: Facilitating Color Authoring for Motion Graphics Video. *In Proceedings of the ACM Human Factors in Computing Systems (CHI). 2024.* DOI
Best Paper Award.

[P5] CHI 2024 **Xinyu Shi**, Mingyu Liu, Ziqi Zhou, Ali Neshati, Ryan Rossi, Jian Zhao. Exploring Interactive Color Palettes for Abstraction-Driven Exploratory Image Colorization. *In Proceedings of the ACM Human Factors in Computing Systems (CHI). 2024.* DOI

[P4] CHI 2023 **Xinyu Shi**, Ziqi Zhou, Julia Zhang, Ali Neshati, Anjul Kumar Tyagi, Ryan Rossi, Shunan Guo, Fan Du, Jian Zhao. De-Stijl: Facilitating Graphics Design with Interactive 2D Color Palette Recommendation. *In Proceedings of the ACM Human Factors in Computing Systems (CHI). 2023.* DOI

[P3] MM 2022 Chaofeng Chen*, **Xinyu Shi***, Yipeng Qin, Xiaoming Li, Xiaoguang Han, Tao Yang, Shihui Guo. Real-World Blind Super-Resolution via Feature Matching with Implicit High-Resolution Priors. *In Proceedings of the ACM Multimedia (MM). 2022.* DOI
Oral Presentation (Top 5%).

[P2] IMWUT 2020
Qijia Shao, Amy Sniffen, Julien Blanchet, Megan E. Hillis, **Xinyu Shi**, Themistoklis Haris, Jason Liu, Melissa Malzkuhn, Lorna Quandt, James Mahoney, David J.M. Kraemer, Xia Zhou, Devin Balkcom. Teaching American Sign Language in Mixed Reality. *In Proceedings of the ACM Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT, presented in Ubicomp 2021).* DOI

[P1] ISMAR 2019
Xinyu Shi, Junjun Pan, Zeyong Hu, Juncong Lin, Shihui Guo, Minghong Liao, Ye Pan, Ligang Liu. Accurate and Fast Classification of Foot Gestures for Virtual Locomotion. *In Proc. of the IEEE Intl. Symposium on Mixed and Augmented Reality (ISMAR). 2019.* DOI

IN SUBMISSIONS

[S2] Under Review
Xinyu Shi, Chun Ye, Shunan Guo, Jane Hoffswell, Gromit Chan, Eunyee Koh, Jian Zhao. Diagrammatic Sketching: Exploring Infographic Design Alternatives. *In Submission.*

[S1] Under Review
Cara Li, Xiang Li, **Xinyu Shi**, Jian Zhao. Semantic Ink: Supporting Reuse with Variation in Annotational Graphics Authoring. *In Submission.*

EXTENDED ABSTRACTS, WORKSHOPS

[A1] CHI EA 2025
Xinyu Shi, Shunan Guo, Jane Hoffswell, Gromit Yeuk-Yin Chan, Victor S. Bursztyn, Jian Zhao, Eunyee Koh. Comprehensive Sketching: Exploring Infographic Design Alternatives in Parallel. *Extended Abstracts of ACM Human Factors in Computing Systems (CHI Late-Breaking Work 2025).* DOI

PAST RESEARCH POSITIONS

2021/09 – Present
University of Waterloo, Waterloo, ON, Canada
Graduate Student Researcher
Advisor: Jian Zhao
Building interactive systems to augment human creativity.

2020/12 – 2021/08
The Chinese University of Hong Kong, Shenzhen, Shenzhen, China
Research Assistant
Advisor: Xiaoguang Han
Research on Image Super-Resolution with VQ-GAN.

2019/09 – 2020/03
Dartmouth College, Hanover, NH, USA
Research Assistant
Advisors: Xia Zhou, Xing-Dong Yang
Research on wearable devices & Mixed Reality for Sign-Language teaching.

2017/10 – 2019/09
Xiamen University, Xiamen, China
Undergraduate Research Assistant
Advisor: Shihui Guo
Research on wearable devices for VR locomotion with gesture recognition.

ACADEMIC SERVICES

Organizing Committee

2025 – 2026 Assistant to Paper Chairs, CHI 2026

Program Committee

2026 Associate Chair, CHI 2027

2025 International Program Committee (Associate Chair), GI 2025

2025 Associate Chair, C&C 2025

Conference Session Chair

2026 Paper Session "*Role Play, Creativity and AI*", CHI 2026

2025 Paper Session "*Malleable and Adaptive Interface*", CHI 2025

Student Volunteer

2026 Graphics Interface (GI)

Reviewer

Received 5 Special Recognitions for Outstanding Reviews

2024 – 2026 ACM User Interface Software and Technology Symposium (UIST)

2024 – 2025 SIGCHI Conference on Human Factors in Computing Systems (CHI)

2026 IEEE Transactions on Visualization and Computer Graphics (TVCG)

2024 – 2025 IEEE Visualization and Visual Analytics Conference (VIS)

2024 – 2025 SIGCHI Late Breaking Work (CHI LBW)

2025 Pacific Conference on Computer Graphics and Applications (Pacific Graphics)

2024 ACM Conference on Designing Interactive Systems (DIS)

2024 Creativity & Cognition (C&C)

INVITED TALKS

2026/05 Guest Lecture for CS449 at the University of Waterloo, hosted by Prof. Jian Zhao.

2026/01 Seminar at the CUHK (Shenzhen), School of AI, hosted by Prof. Feng Yin.

2025/11 University of Toronto iSchool Seminar, hosted by Prof. Ebrahim Bagheri.

2025/11 Guest Lecture for CS889 at the University of Waterloo, hosted by Prof. Jian Zhao.

2025/11 UCSD Design Lab Seminar, hosted by Sirui Tao and Zhicheng Huang.

2025/09 Invited Talk at Adobe (MOVIN Group), hosted by Rubaiat Habib Kazi.

MENTORSHIP

2025 – **Cara Li**, Master student at University of Waterloo.
Project: Fluid in-situ graphics creation through reified sketched annotations [S1].

2024 – 2025 **Chun Daisy Ye**, Undergraduate student at University of Waterloo, published [S2].
Now: Software Engineer at Salesforce.

2023 – 2024 **Yinghou Wang**, Master student at Harvard University & MIT, published [P6, P7].
Now: Creative Technologist at Motorola Solutions.

2023 **Mingyu Liu**, Undergraduate student at University of Waterloo, published [P5].
Now: Software Engineer at Bloomberg.

2022 **Ziqi Zhou**, Master student at University of Waterloo, published [P4].
Now: Partner at Puzzle Ventures.

2022 **Julia Zhang**, Undergraduate student at University of Waterloo, published [P4].
Now: Software Engineer at Whatnot.

TEACHING EXPERIENCE

Teaching Assistant in University of Waterloo

CS246 Object-Oriented Software Development. (Winter 2022, Fall 2023, Winter 2025)

CS251 Computer Organization and Design. (Spring 2022, Winter 2023)

CS234 Data Types and Structures. (Fall 2022)

CS115 Introduction to Computer Science. (Fall 2021)

PRESS COVERAGE

2024 *Waterloo News*: Scheming colours faster and easier. [Link]

2023 *GraphicSpeak*: University of Waterloo's Computer Scientists want Graphic Designers to use better Colors. [Link]

2023 *TechTimes*: Computer Scientists Develop "De-Stijl" Tool With Adobe to Help People Use Color Better in Graphic Design. [Link]

2023 *Waterloo News*: Dreaming in technicolour: Waterloo computer scientists develop a tool to help people use colour better in graphic design. [Link]